iv) Gear efficiency is a measure of:

a) The smoothness of gear operation
b) The ratio of input power to output power
c) The durability of the gears
d) The size of the gears

v) How are cobots programmed for different tasks?

a) They learn on their own through trial and error.
b) They require complex programming by robotics experts.
c) They can be easily programmed by non-experts using intuitive interfaces.
d) They cannot be programmed once their initial tasks are defined.

vi) Robot is derived from Czech word.

a) Rabota b) Robota c) Rebota d) Ribota

ix) Which type of sensor is commonly used to detect the presence of objects without physical contact.

(a) Proximity sensor (b) Tectile sensor (c) Temperature sensor (d) Accelerometer

x) Which sensor is commonly used in robotics to detect touch or pressure?

(a) Proximity sensor (b) Tactile sensor (c) Gyroscope (d) Microphone

xi)_____ provide information about the systems interaction with its environment?

a) External sensors b) internal sensor c) thermal sensors d) environmental sensor

xviii) This list contains items of what data type?

list = [1,4,6]

a) String b) Boolean c) Float d) Integer

i) Define the role of actuators in robotics with some key reasons **[3]**
ii) Describe Gear Ratio with an example **[3]**

iii) Write a python program that perform the following operations on a list of integers. **[8]**

1) Create a list of the integers: [17,85,74,28,291,175]
2) Sort the list in ascending order.
3) Reverse the list.
4) Append integer 89 to the list.
5) Insert integer 19 at index 5
6) Delete integer 291 from the list.
7) Print the final list.

ii) Which one of the following statements about cobots is true?

a) Cobots eliminate the need for human presence

b) Cobots work collaboratively with humans

c) Cobots are used only in educational settings

d) Cobots are outdated technology

iii) Which of the following is an example of a robotic exoskeleton used in rehabilitation?

a) Robotic vacuum cleaner
b) Robotic prosthetic arm
c) Robotic pet companion
d) Robotic lawn mower

iv) Camera is an example of…

a) Pressure Sensor
b) Vision Sensor
c) Solar Sensor
d) Motion Sensor

v) How are cobots programmed for different tasks?

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i) Explain the importance of integrating sensors, actuators and controllers in a robotics system. **[3]**

[3]

iii) Write a python program that performs the following operations on a tuple. **[9]**

1) Create a tuple with the elements (5,10,15,20,25)
2) Access and print the element at index 2.
3) Convert the tuple into a list.
4) Append the integer 30 to the list.
5) Convert the list back into a tuple.
6) Print the final tuple

(x) What will be the output of the following code?

```
my_list = [1, 2, 3, 4, 5]
new_list = my_list[1:4]
new_list.append(6)
print(my_list)
```

(a) [1, 2, 3, 4, 5, 6] (b) [1, 2, 3, 4, 5] (c) [1, 2, 6, 3, 4, 5] (d) [1, 2, 3, 4, 5] (no change)

(xviii) Which sensor is used by robots to measure changes in motion or orientation?
(a) Gyroscope Sensor (b) Pressure Sensor (c) Ultrasonic Sensor (d) Camera Sensor

(iii) What are the key components in the block diagram of a basic robot and how do they work together? **[2]**

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(iii) Write a Python program that performs the following operations on a list of mixed data types (integers, strings, and floats): **[9]**

1. Create a list: [100, "apple", 200, "banana", 300.5].
2. Add the float 400.75 to the list.
3. Insert the string "orange" at index 2.
4. Remove the string "banana".
5. Print the length of the list.
6. Print the final list.

(iii) Write a Python program that performs the following operations on a tuple of city names: **[9]**

1. Create a tuple with the elements: ("New York", "London", "Tokyo", "Paris").
2. Access and print the last element.
3. Convert the tuple into a list.
4. Insert the city "Dubai" at index 2.
5. Convert the list back into a tuple.
6. Print the final tuple.